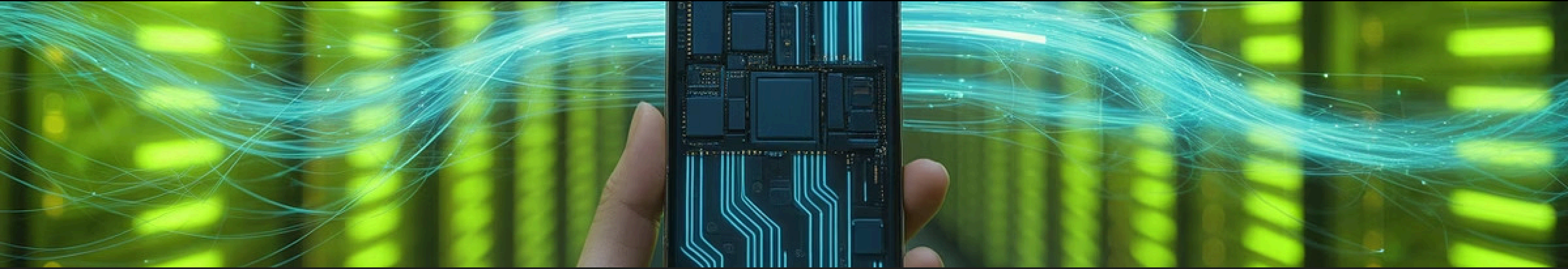
Advanced Data Pipeline and Transformation Using Azure

Welcome to Semester 5's most exciting journey! This isn't just another lecture – it's your launchpad into the world of modern data engineering. Over the next 120 minutes, we'll explore how data flows through the digital ecosystem that powers every app you use, every recommendation you receive, and every transaction you make.



Your Data in Motion 📱

Quick Activity

Take 2 minutes with your neighbour and discuss: **What was the last app you used today? Where do you think its data went?**

Think about the journey from your phone tap to the cloud servers, databases, and analytics engines working behind the scenes.

The Data Journey

Every interaction you have creates a data trail that gets processed, transformed, and analysed in real-time. Understanding this flow is the foundation of modern data engineering.

Data is Everywhere 🌍

Swiggy Order Tracking

Your order status updates in real-time as it moves from restaurant to your doorstep, powered by location data pipelines and event streaming.

Netflix Recommendations

Machine learning algorithms process your viewing history, time spent watching, and global trends to suggest your next binge-worthy series.

Paytm Payments

Every transaction triggers fraud detection algorithms, balance updates, and merchant analytics – all happening in milliseconds.

Quick Poll: Which of these data experiences do you relate to the most? How often do you think about the complex data systems running behind these simple interactions?

Why This Subject? 🎯

Market Demand

Data engineering roles have grown by 35% year-over-year, making it one of the fastest-growing tech careers in India and globally.

Salary Potential

Entry-level data engineers earn ₹8-12 LPA, while senior professionals command ₹25-40 LPA at top companies.

Future-Proof Skills

With AI and ML becoming mainstream, the demand for professionals who can build reliable data infrastructure will only increase.

The Course Journey Map 🛣️

1

Unit 1: Pipeline Foundations

Learn the building blocks of data movement and transformation

2

Unit 2: Real-Time Processing

Master streaming data and event-driven architectures

3

Unit 3: Big Data with Databricks

Scale your solutions for massive datasets

4

Unit 4: Governance & Security

Ensure data quality and compliance

5

Unit 5: Optimization

Make your solutions cost-effective and performant

6

Unit 6: DevOps Automation

Deploy and maintain production-ready systems

We'll progress step by step, building on each unit's concepts to create comprehensive data engineering expertise. By the end, you'll have the skills to design enterprise-grade data systems.

CLOs in Simple Words 🧩

01

Build

Create robust data pipelines that can handle various data sources and formats

02

Real-Time Processing

Implement streaming solutions for live data analysis and immediate insights

03

Transform

Clean, enrich, and reshape data to make it analysis-ready

04

Govern

Ensure data quality, security, and compliance across all systems

05

Optimize

Fine-tune performance and manage costs effectively

06

Automate

Deploy CI/CD practices for reliable, scalable data operations

Activity: Guess the Unit 🤔

Unit 2 Challenge: Real-time Data Processing

Question: "Where have you experienced real-time data processing in your daily life?"

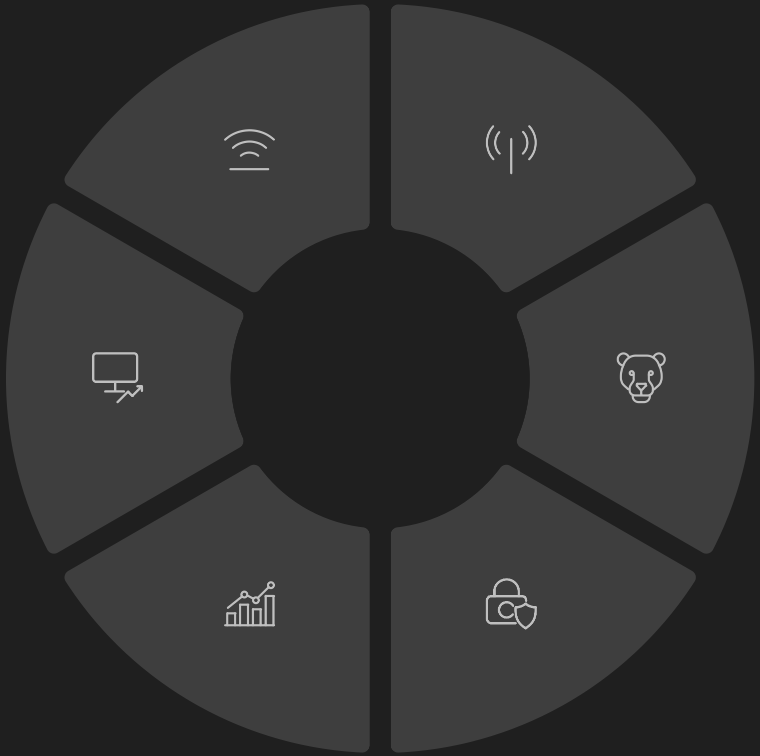
Expected Examples:

- Spotify's live lyrics and recommendations
- Instagram Live viewer counts
- UPI fraud detection during payments
- Google Maps traffic updates
- Stock market price updates

The Technology Behind It:

These experiences are powered by stream processing engines that can handle thousands of events per second, making split-second decisions that enhance your user experience.

Unit Snapshots 📖



Unit 1: Pipelines

Foundation of data movement and ETL processes

Unit 2: Real-time

Event-driven architectures and stream processing

Unit 3: Big Data

Databricks and distributed computing

Unit 4: Governance

Data quality, security, and compliance

Unit 5: Optimization

Performance tuning and cost management

Unit 6: DevOps

CI/CD and infrastructure automation

Tools You'll Master 🔧

Azure Data Factory

Your pipeline orchestration powerhouse

Stream Analytics

Real-time event processing engine

Databricks

Unified analytics platform for big data

Azure Purview

Data governance and cataloguing

Azure Monitor

Performance monitoring and alerting

Azure DevOps

CI/CD and project management

Quick Poll: Which of these tools have you already heard of or used? Don't worry if they're all new – we'll master them together!

Hands-On Skills 💻

ADF Pipeline Design

Drag-and-drop interface for creating complex data workflows with error handling and monitoring

PySpark Transformations

Write efficient Python code for distributed data processing on massive datasets

Stream Analytics Queries

SQL-like syntax for real-time data filtering, aggregation, and pattern detection

Data Governance

Implement lineage tracking, quality rules, and compliance frameworks

DevOps Pipeline Automation

Set up continuous integration and deployment for data infrastructure

Mini Projects Preview 🌟

Retail Real-Time Dashboard

Build a live analytics dashboard showing sales trends, inventory levels, and customer behaviour as it happens. Perfect for understanding streaming data concepts.

End-to-End Data Lake System

Create a complete data lake solution from raw data ingestion to business intelligence reporting, covering the entire data engineering lifecycle.

Governance Automation Tool

Develop automated data quality checks and compliance reporting systems that ensure your data meets industry standards.

Cost-Optimized Pipelines

Design and implement data pipelines that balance performance with cost-effectiveness, learning crucial optimization techniques.

CI/CD for Data Pipelines

Set up automated testing and deployment workflows for data infrastructure, bringing DevOps practices to data engineering.

Discussion Question: Which project excites you the most? Think about which one aligns best with your career interests.

Activity: Pick Your Project Team Idea 🤝

Team Formation Activity

Time: 5 minutes

Group Size: 4-5 students

Your Mission:

Form teams and discuss: **"If your team had to pick one mini-project to work on this semester, which would you choose and why?"**

Consider:

- Team members' interests
- Skill development goals
- Industry relevance
- Learning complexity

Expected Outcome:

Each team should have a preliminary project preference and understand what excites them most about data engineering.

Don't worry – you'll have time to change your mind as we dive deeper into each unit!

Real-World Examples 🌐

Netflix Recommendations

Processes 500+ billion events daily to power their recommendation engine, using real-time streaming and machine learning pipelines.

Uber Live Tracking

Handles millions of location updates per minute, matching riders with drivers using geospatial data processing and real-time analytics.

Zomato Surge Pricing

Analyses demand patterns, weather data, and delivery capacity in real-time to optimize pricing and delivery routes.

These companies rely on the exact same Azure tools and techniques you'll be learning. The skills you develop here directly translate to solving real business problems at scale.

Activity: Spot the Pipeline 🔍

1

Bank Fraud Detection

A customer makes a transaction that gets flagged as potentially fraudulent within 100 milliseconds. The system blocks the transaction and sends an SMS alert.

Your Task: Which unit would handle this scenario?

2

YouTube Trending

YouTube processes billions of views, likes, and shares to determine trending videos. The trending page updates every few minutes with fresh content.

Your Task: What kind of data processing is this?

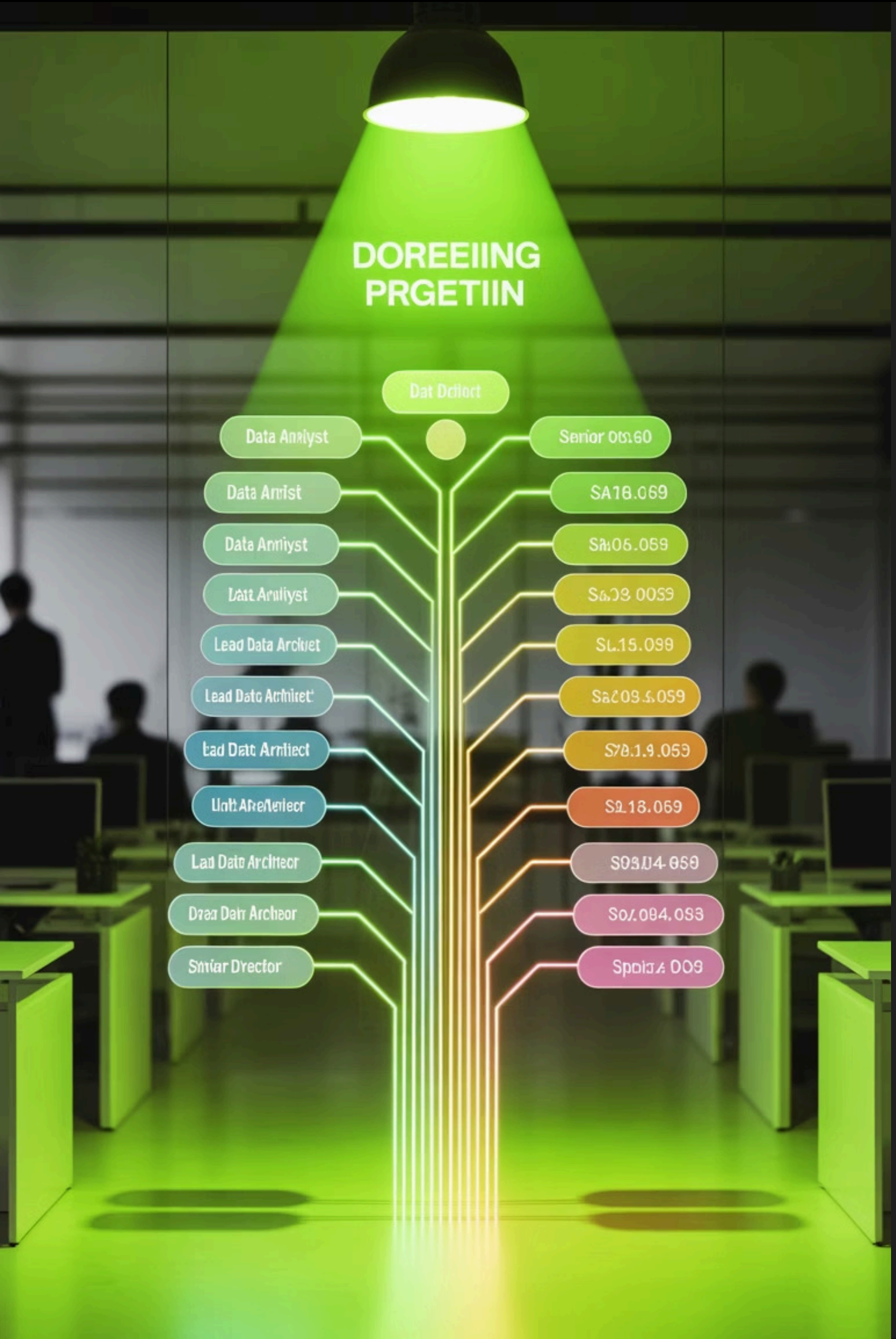
3

Flipkart Delivery Tracking

From order placement to doorstep delivery, customers can track their packages in real-time, receiving updates at every checkpoint.

Your Task: Identify the data pipeline components.

Interactive Poll: We'll vote on the answers together and discuss the underlying technology!



Skills → Careers 💼

Data Engineer

₹8-15 LPA entry level | Build and maintain data pipelines, ETL processes, and data infrastructure

Streaming Data Analyst

₹12-20 LPA | Specialize in real-time analytics, event processing, and streaming architectures

Data Governance Specialist

₹15-25 LPA | Focus on data quality, compliance, security, and enterprise data management

Senior Cloud Data Engineer

₹20-35 LPA | Lead complex data projects, mentor teams, design enterprise architectures

Each unit in this course directly maps to these career paths, giving you flexibility to specialize in areas that match your interests and strengths.

Class Engagement Plan 📅

Pair Programming

Work together on coding challenges, share knowledge, and learn from each other's approaches to problem-solving.

Real-Time Quizzes

Interactive polls and quick knowledge checks using Kahoot or similar tools to keep everyone engaged and assess understanding.

Case Studies

Analyse real-world scenarios from companies like Swiggy, PayTM, and Flipkart to understand practical applications.

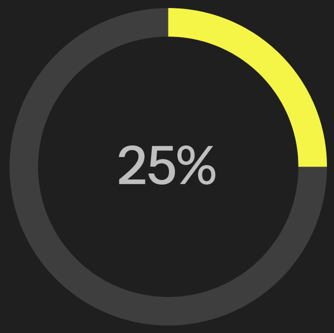
Group Projects

Collaborative mini-projects that simulate real workplace scenarios and build teamwork skills alongside technical expertise.

This isn't a traditional lecture-based course. Expect 70% hands-on activities, 20% interactive discussions, and 10% traditional teaching.

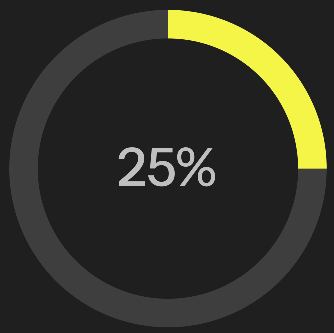
Assessment & BYOD 📝

Assessment Breakdown



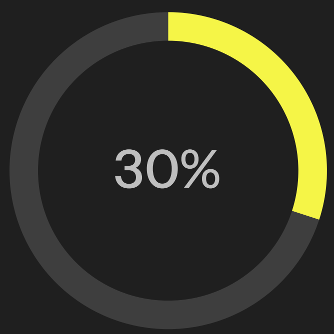
Continuous Assessment (CA)

Regular quizzes, assignments, and participation



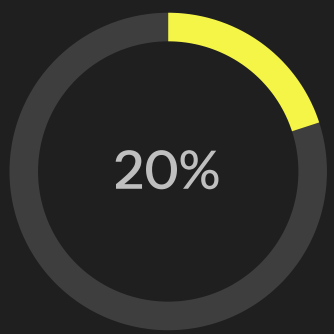
Mid-Term Test (MTT)

Practical examination covering Units 1-3



End-Term Test (ETT)

Comprehensive practical and theory examination



Projects

Mini-projects and final capstone project

BYOD Requirements

Bring Your Own Device

- Laptop with 8GB+ RAM
- Reliable internet connection
- Azure account (we'll help set up)
- Python development environment

40-50% hands-on emphasis: Most assessments will be practical, not theoretical memorization.

Expectation Setting 📌

What We Expect From You

- **Be Curious:** Ask questions, experiment with tools, and explore beyond the syllabus
- **Participate Actively:** Join discussions, collaborate in teams, and share your insights
- **Experiment Fearlessly:** Don't be afraid to break things – that's how you learn best
- **Stay Consistent:** Regular practice is key to mastering data engineering concepts

What You Can Expect From Us

- **Hands-On Learning:** 70% practical activities with real-world tools and scenarios
- **Industry Context:** Every concept tied to actual use cases from leading tech companies
- **Personal Guidance:** Individual attention for projects and career development
- **Current Content:** Up-to-date curriculum reflecting latest industry practices

Reflection Activity 🪞

One Word Reflection

Poll Question: "One word to describe how you feel about this subject right now."

Possible Responses:

- Excited 🚀
- Curious 🤔
- Overwhelmed 😅
- Confident 💪
- Motivated 🔥
- Uncertain ❓
- Ready 🎯

Activity Instructions:

Use post-it notes, chat responses, or our polling tool to share your word. We'll create a live word cloud together!

Remember, all feelings are valid – whether you're excited or nervous, we'll work through this journey together.

Your response helps us understand where you're starting from and how we can best support your learning journey. There are no wrong answers here – just honest reflections!

Ready to Begin? 🚀

Your Data Engineering Journey Starts Now

You've gotten a taste of what's ahead – six units of hands-on learning, real-world projects, and industry-relevant skills. From basic pipelines to advanced DevOps automation, you'll build expertise that directly translates to exciting career opportunities.

Remember: this isn't just about learning tools and technologies. You're preparing to solve real problems for real companies, to make data work better, faster, and smarter.

Stay Curious

Question everything, experiment fearlessly

Think Big

Every pipeline you build could power the next unicorn

Build Together

Your classmates are your future network

Next Session: We dive into Unit 1 with hands-on Azure Data Factory pipeline creation. Come prepared with your laptops and your curiosity!